

Semantic Segmentation

Task definition: Semantic Segmentation



CAT

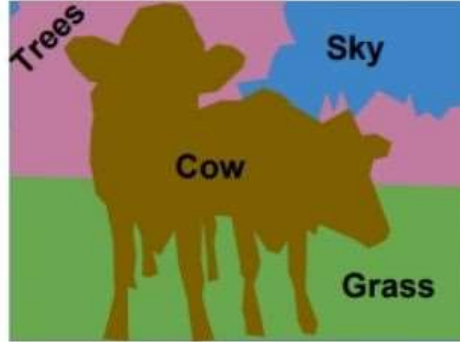
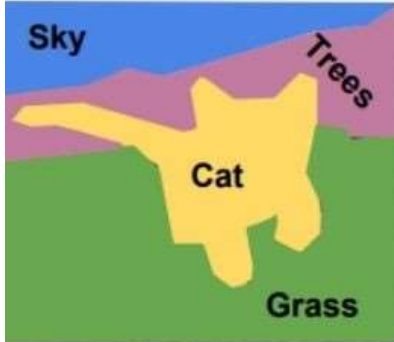
Classify the main object in the image.



CAT, GRASS,
TREE, SKY

No objects, just classify each pixel.

Semantic Segmentation



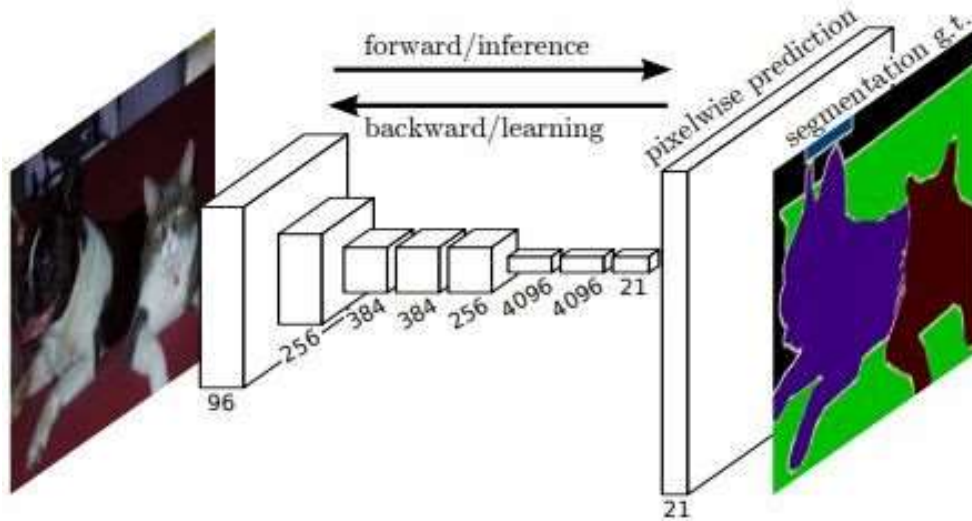
- Every pixel in the image needs to be labelled with a category label.

- Do not differentiate between the instances (see how we do not differentiate between pixels coming from different cows).

Fully Convolutional Networks

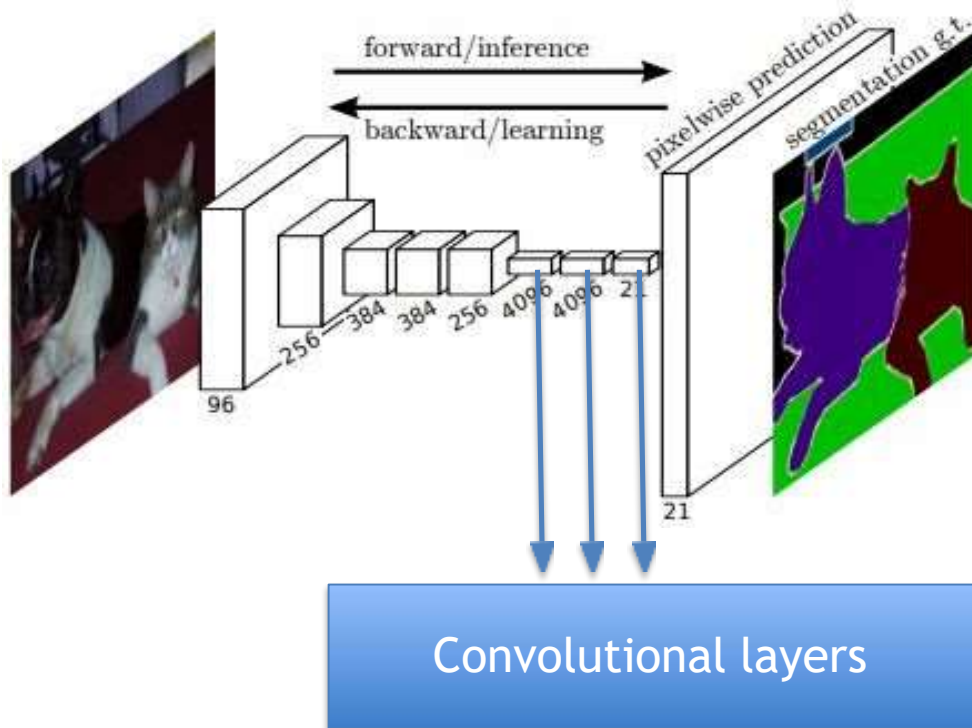
Fully convolutional neural networks

- A FCN is able to deal with any input/output size

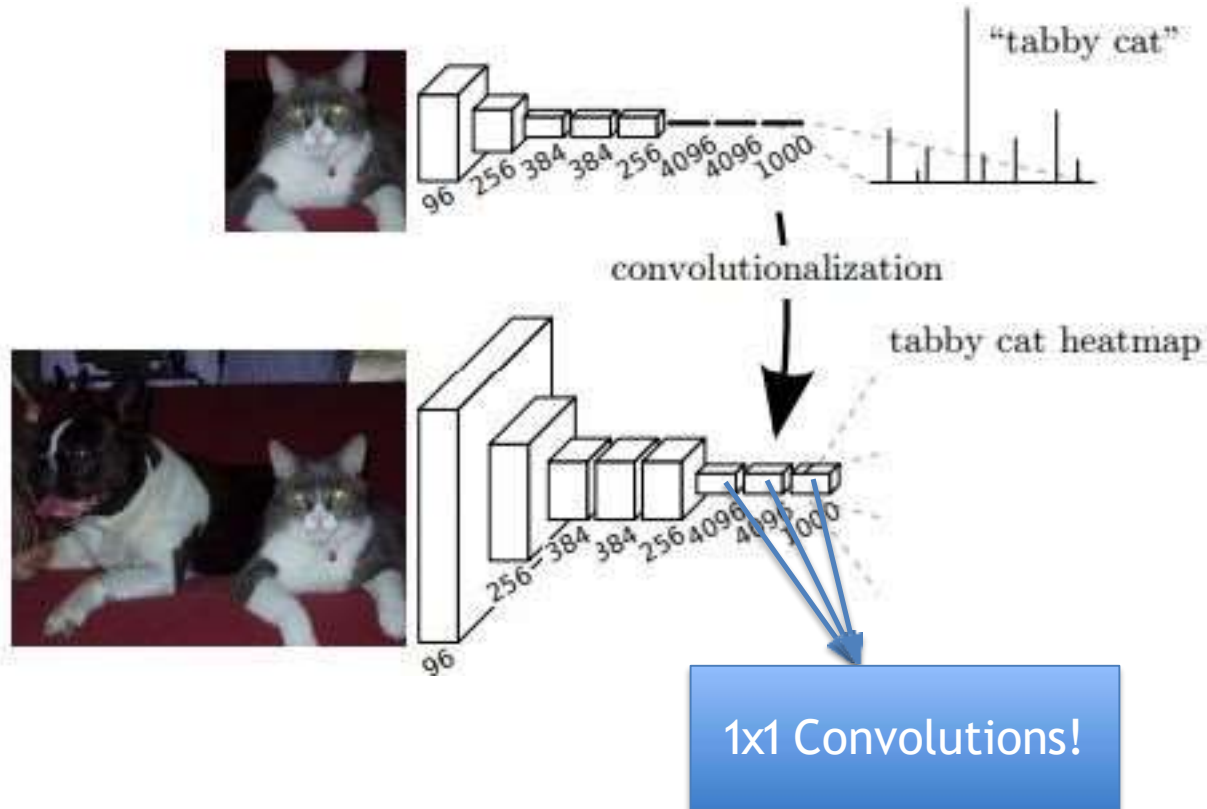


Fully convolutional neural networks

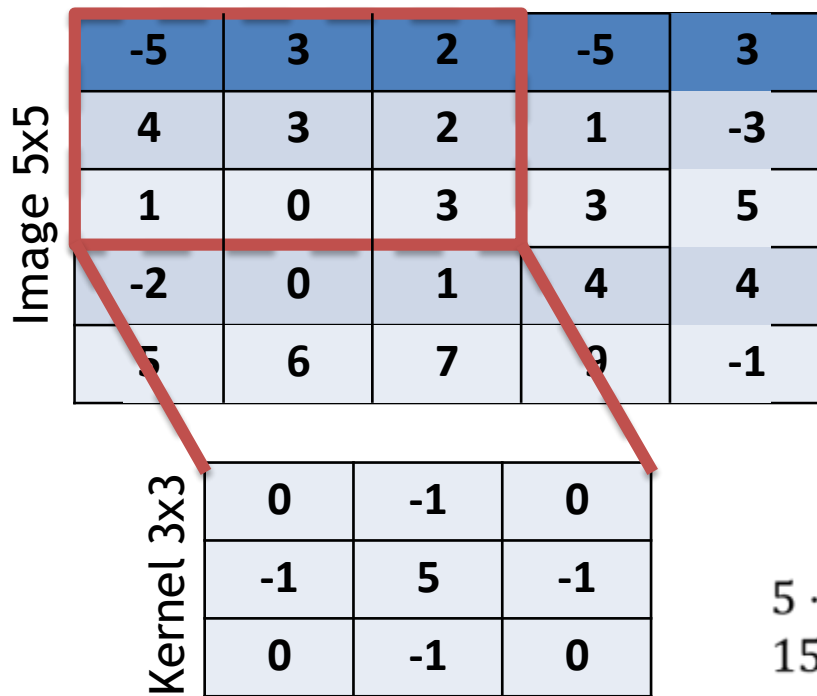
1. Replace FC layers with convolutional layers.
2. Convert the last layer output to the original resolution.
3. Do softmax-cross entropy between the pixelwise predictions and segmentation ground truth.
4. Backprop and SGD



“Convolutionalization”



Recall: Convolutions on Images

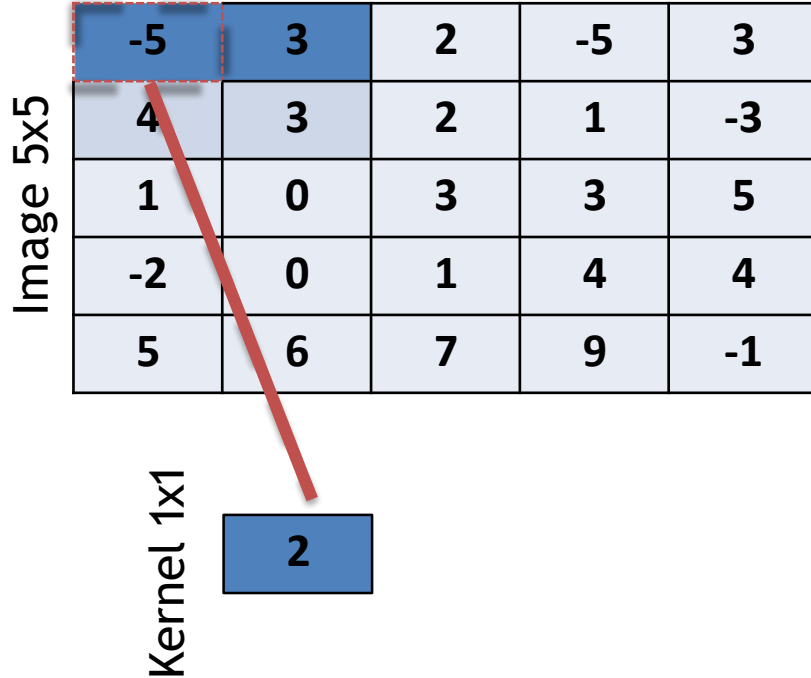


Output 3x3

6		

$$5 \cdot 3 + (-1) \cdot 3 + (-1) \cdot 2 + (-1) \cdot 0 + (-1) \cdot 4 = 15 - 9 = 6$$

1x1 Convolution



What is the output size?

1x1 Convolution

Image 5x5

-5	3	2	-5	3
4	3	2	1	-3
1	0	3	3	5
-2	0	1	4	4
5	6	7	9	-1

Kernel 1x1

2

$$-5 * 2 = -10$$

-10				

1x1 Convolution

Image 5x5

-5	3	2	-5	3
4	3	2	1	-3
1	0	3	3	5
-2	0	1	4	4
5	6	7	9	-1

-10	6	4	-10	6
8	6	4	2	-6
2	0	6	6	10
-4	0	2	8	8
10	12	14	18	-2

Kernel 1x1

2

$$-1 * 2 = -2$$

1x1 Convolution

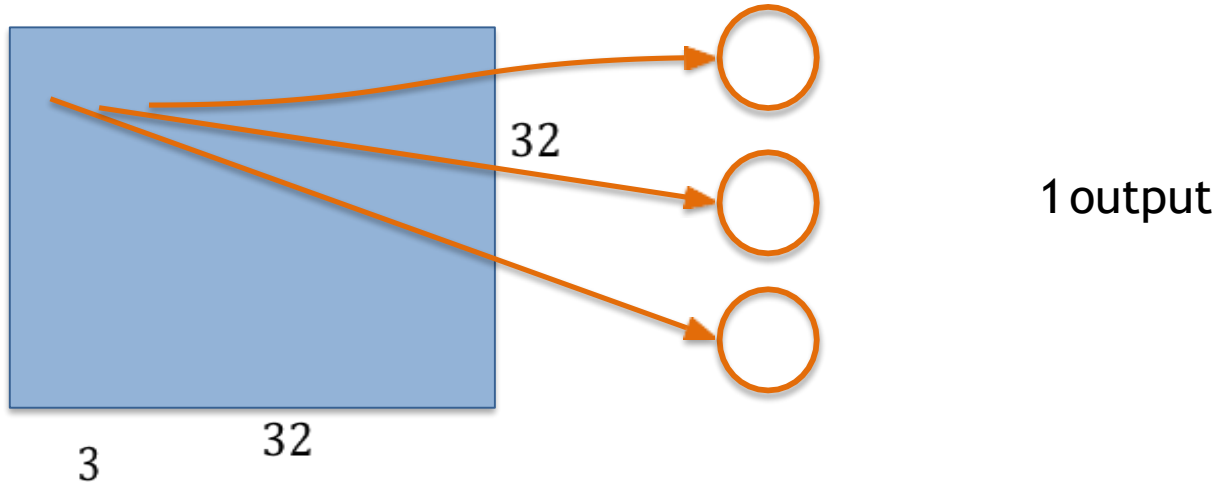
Image 5x5

-5	3	2	-5	3
4	3	2	1	-3
1	0	3	3	5
-2	0	1	4	4
5	6	7	9	-1

-10	6	4	-10	6
8	6	4	2	-6
2	0	6	6	10
-4	0	2	8	8
10	12	14	18	-2

- 1x1 kernel: keeps the dimensions and scales input

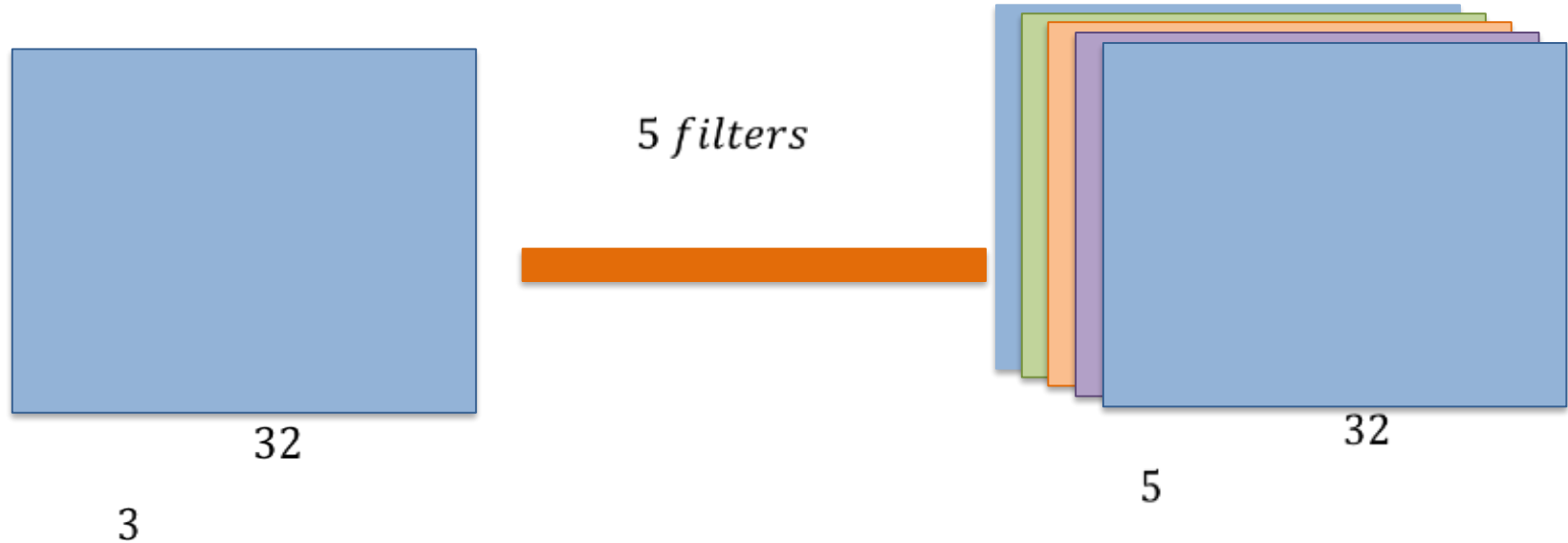
1x1 Convolution



- Same as having a 3 neuron fully connected layer

1x1 Convolution

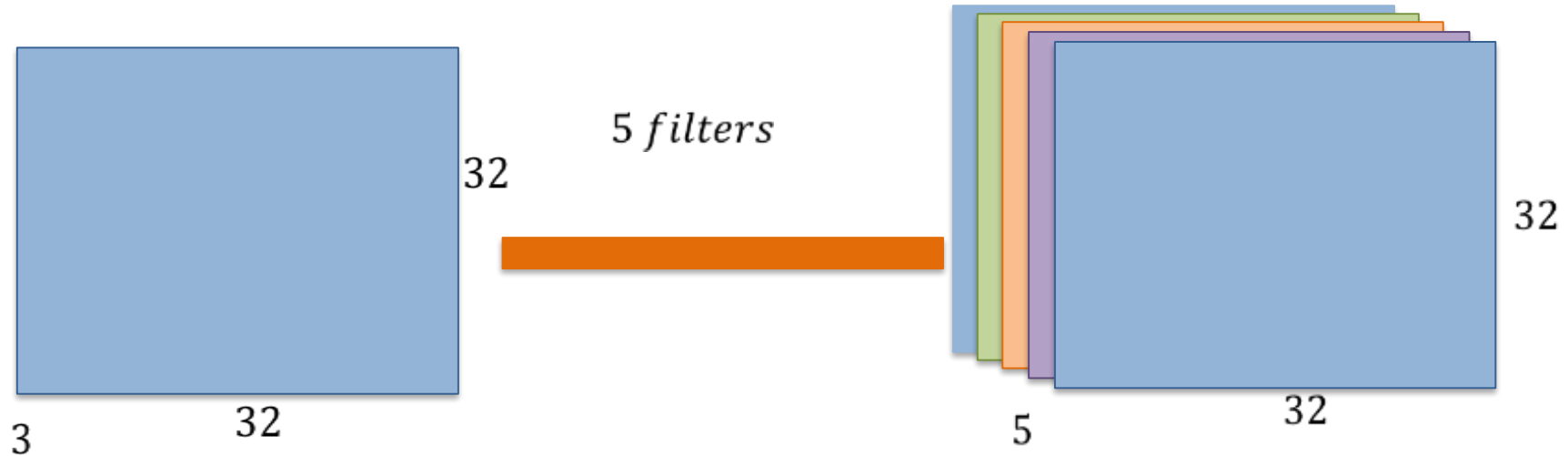
[Li et al. 2013]



- As always we use more convolutional filters

Network in Network

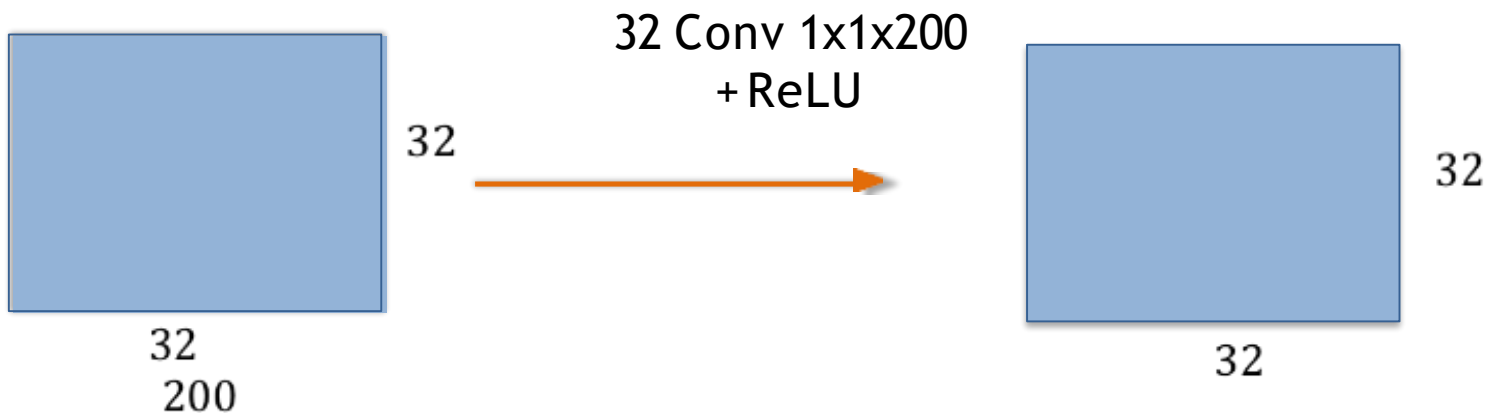
[Li et al. 2013]



- As always, we use more convolutional filters

Using 1x1 Convolutions

- Use it to shrink the number of channels
- Further adds a non-linearity $\square$ one can learn more complex functions



🔗 1×1 Convolution in FCN

- **Input:**

$H \times W \times C$ (feature map)

- **Operation:**

$(H, W, C) \rightarrow (H, W, \text{num_classes})$

□ **Key Idea**

- Each pixel has a **C-dimensional feature vector**
- Converted into **class scores**
- 🖐️ Acts as a **pixel-wise classifier**

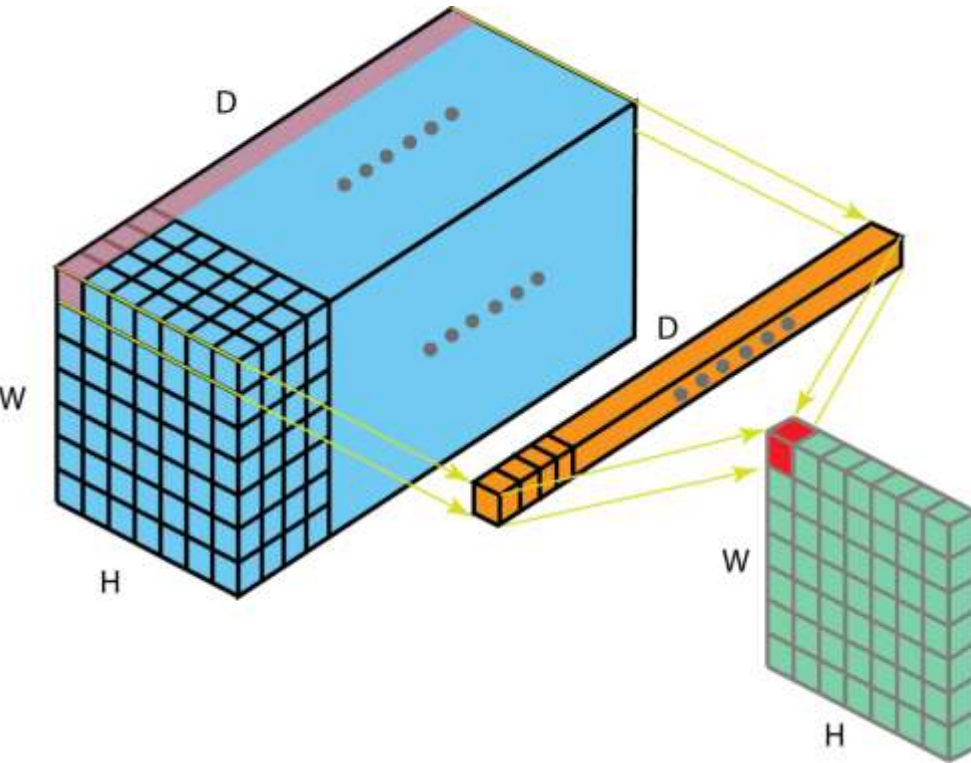
💡 Why 1×1 Conv?

- Preserves **spatial structure**
- Performs **channel-wise linear combination**
- Equivalent to a **fully connected layer per pixel**

🔄 Convolutionalization

- Replace **FC layers** → **1×1 conv**
- Enables **dense prediction (per-pixel classification)**

Using 1x1 Convolutions



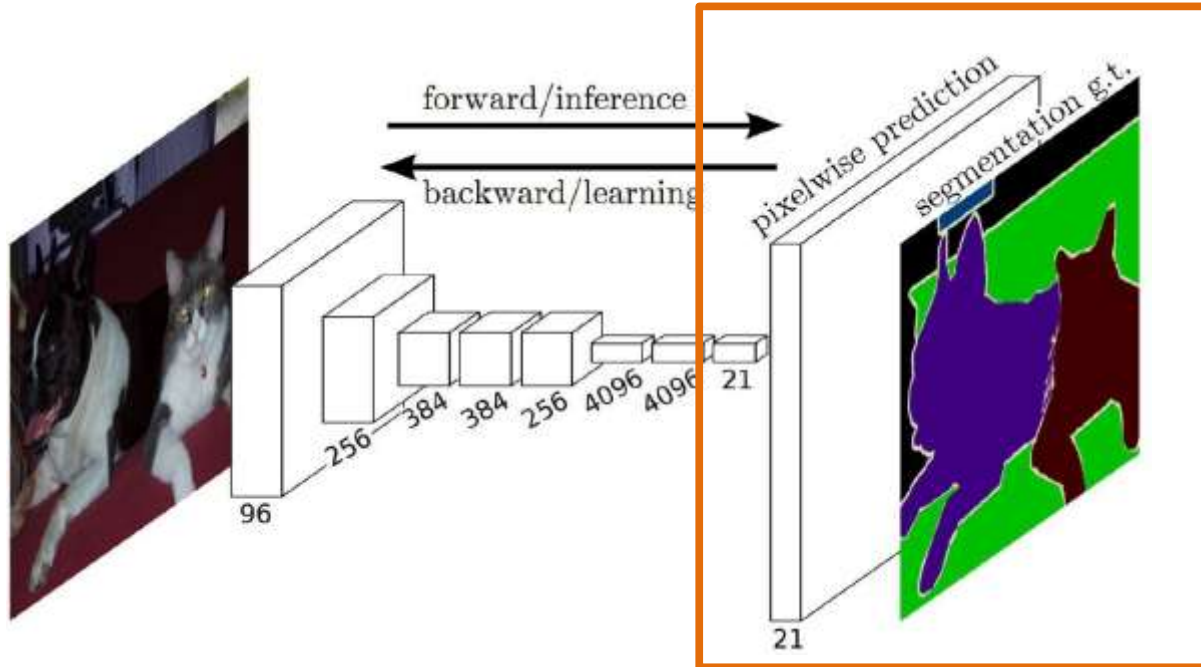
💡 Key Idea:

- Acts as a **pixel-wise classifier**
- Combines channel information **without changing spatial size**
- Equivalent to a **fully connected layer applied at each pixel**

“1×1 convolution turns feature maps into pixel-wise class predictions without affecting spatial resolution.”

Semantic Segmentation (FCN)

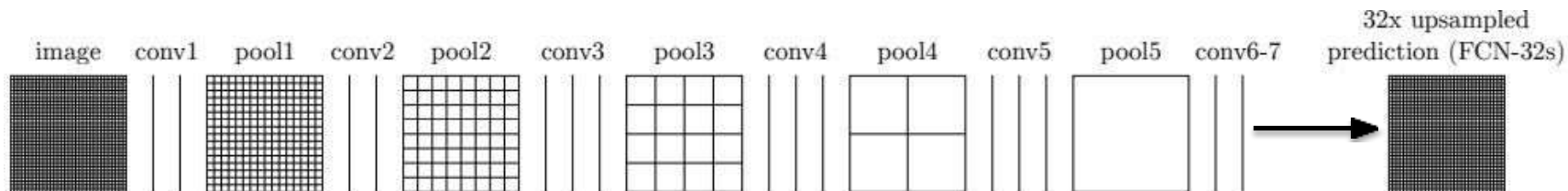
- Fully Convolutional Networks for Semantic Segmentation



How do we
upsample?

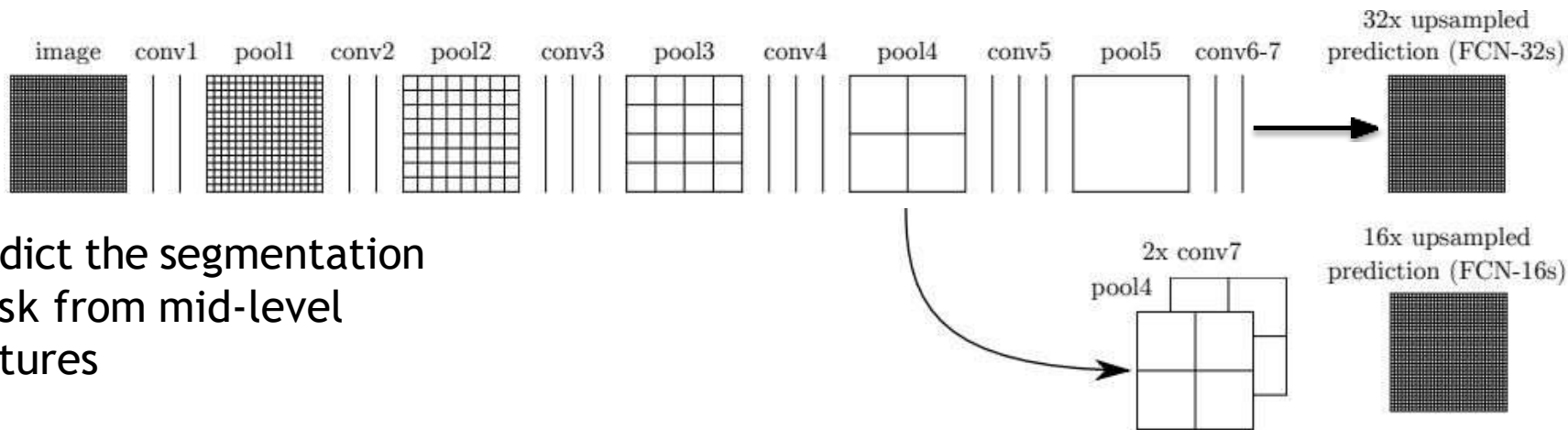
Network's architecture

Predict the
segmentation mask
from high level features



Network's architecture

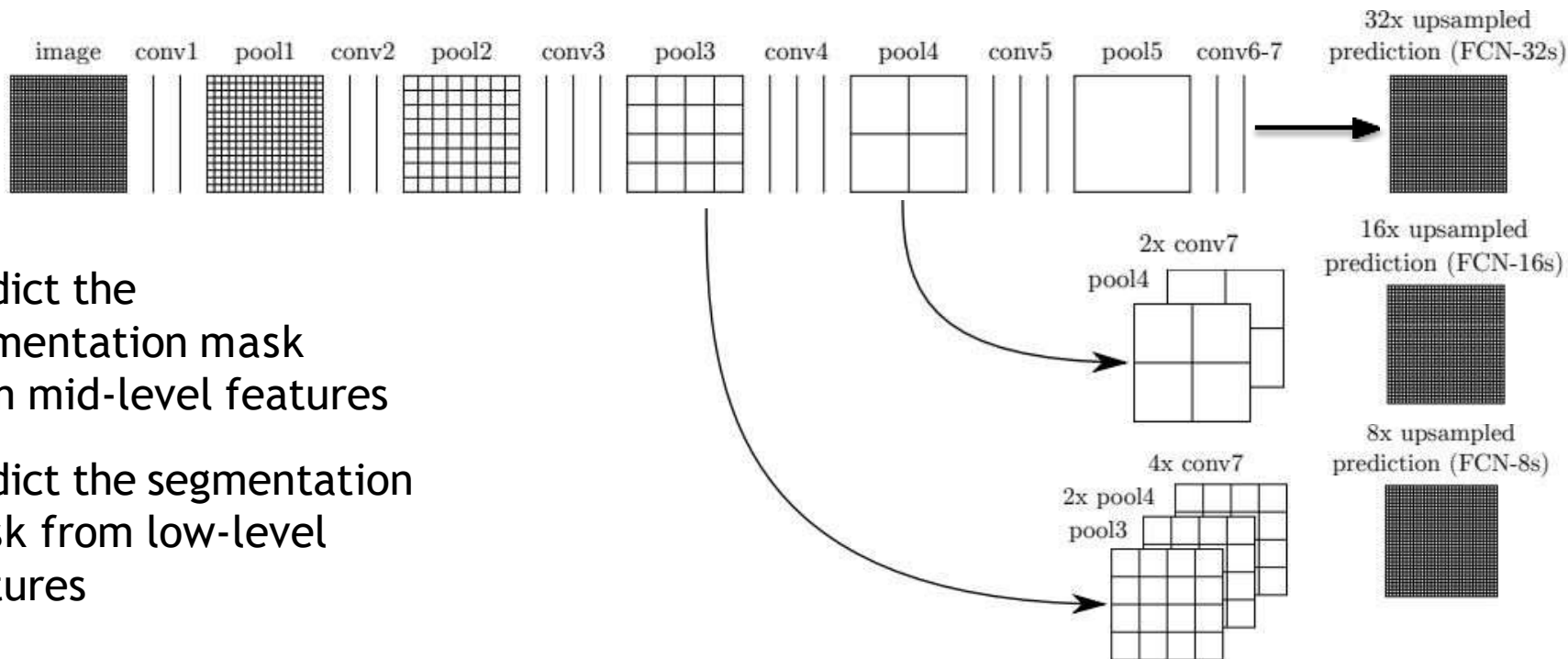
Predict the segmentation mask from high level features



Predict the segmentation mask from mid-level features

Network's architecture

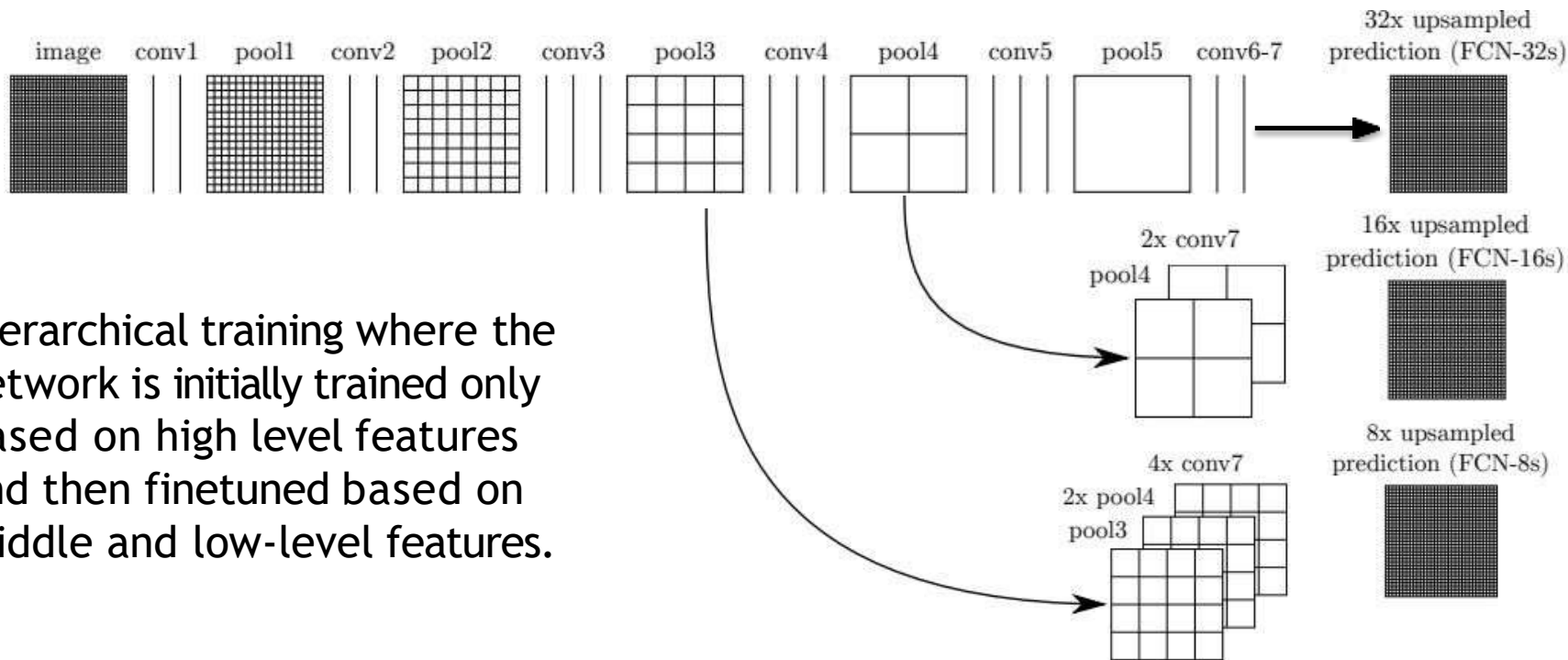
Predict the segmentation mask from high level features



Predict the segmentation mask from mid-level features

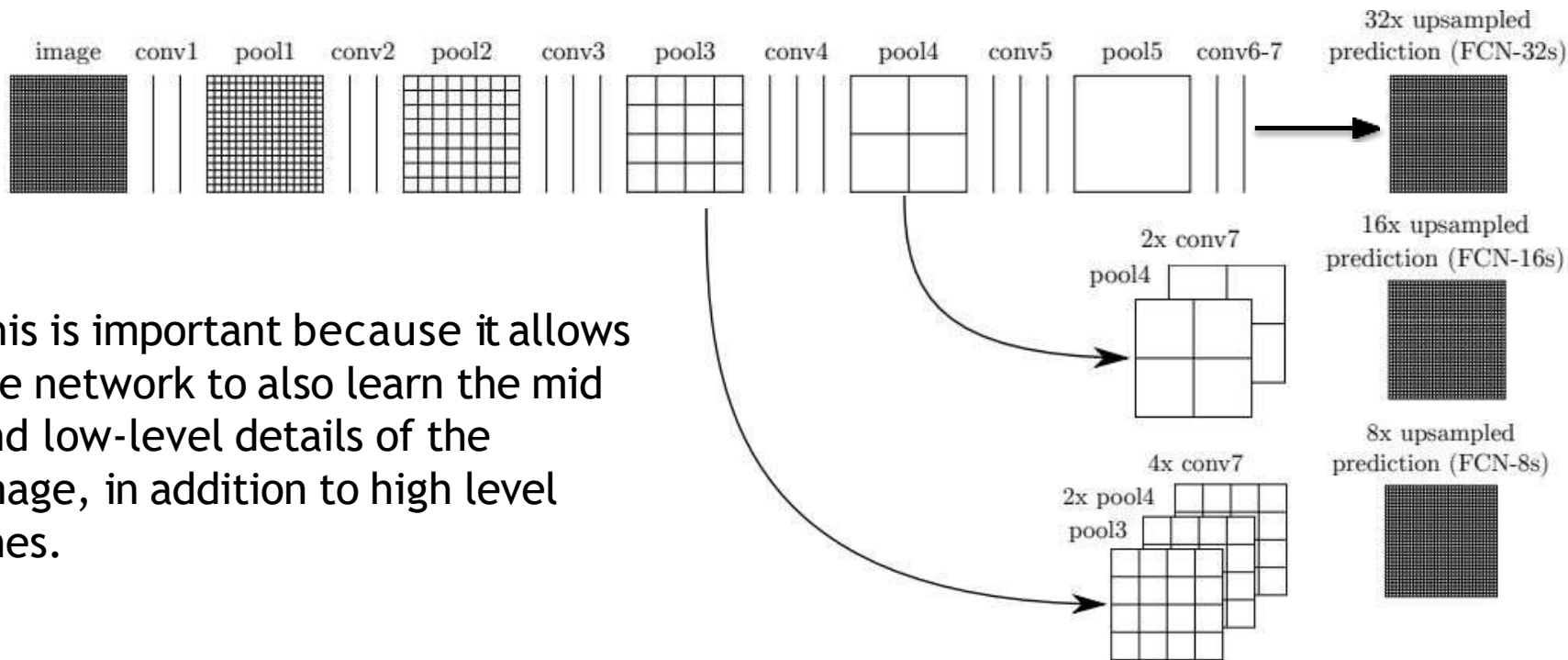
Predict the segmentation mask from low-level features

Network's architecture



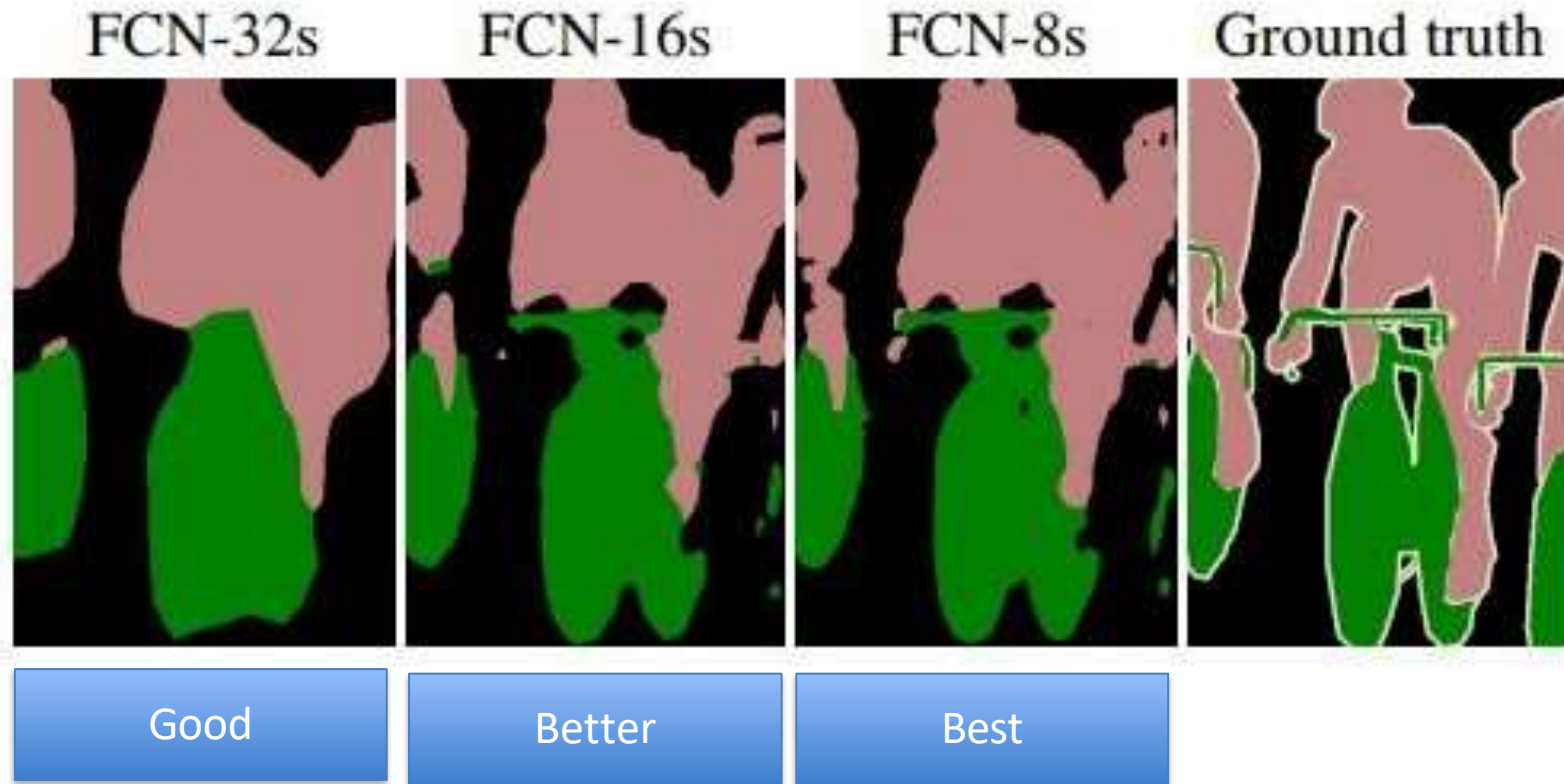
Hierarchical training where the network is initially trained only based on high level features and then finetuned based on middle and low-level features.

Network's architecture

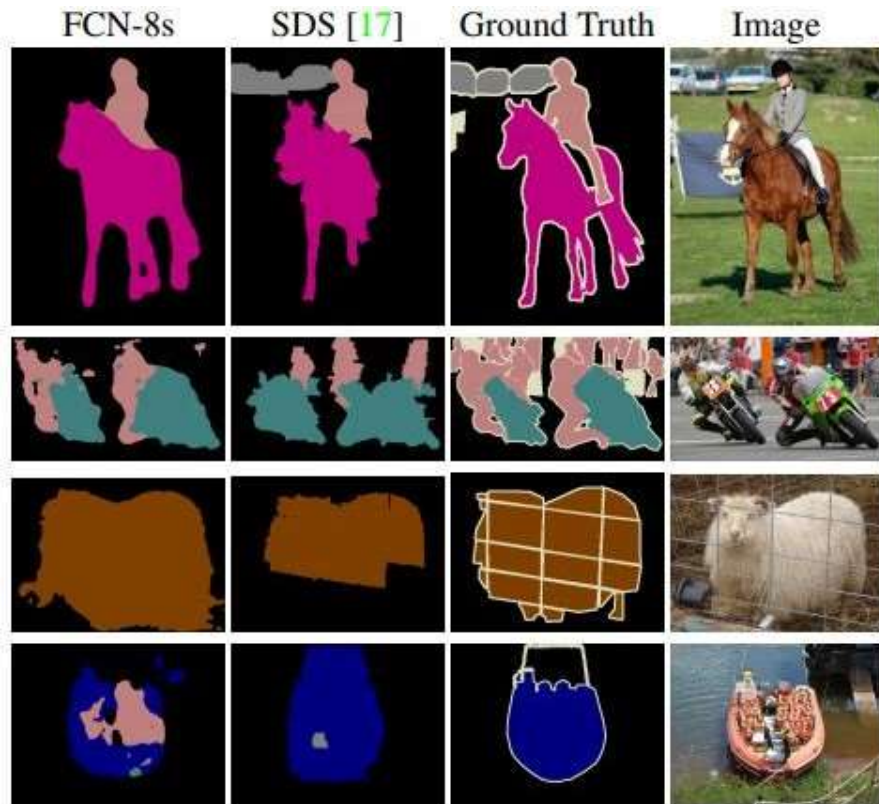


This is important because it allows the network to also learn the mid and low-level details of the image, in addition to high level ones.

Qualitative results



Qualitative results



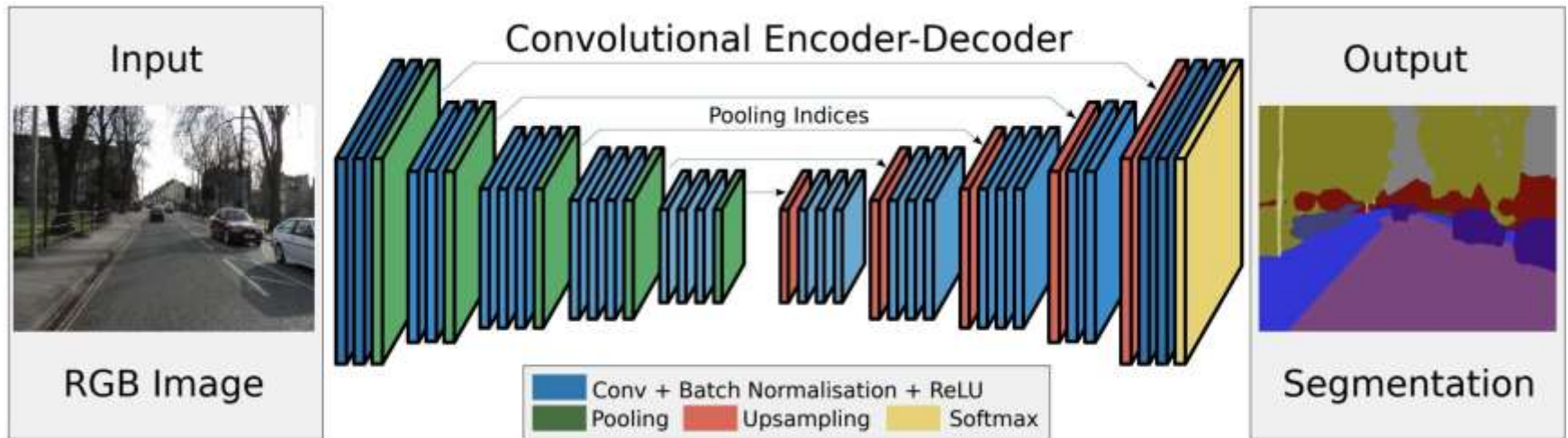
SDS (Simultaneous Detection and Segmentation) is an R-CNN-based method, i.e., it uses object proposals. It was introduced by Ross Girshick et al. (2014).

In general, FCN outperforms significantly (both qualitatively and quantitatively) pre-deep learning and quasi-deep learning methods and is recognized as the AlexNet of semantic segmentation.

Autoencoder-style architecture

SegNet

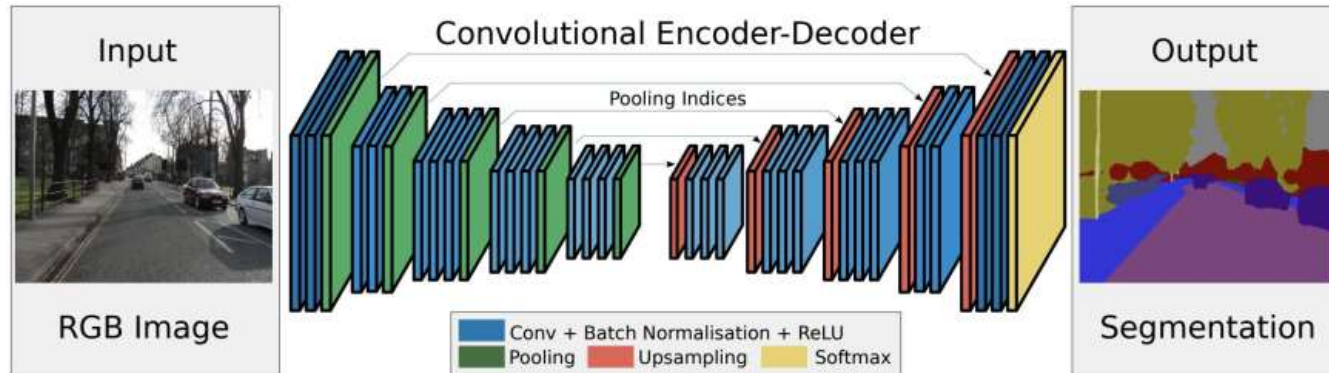
- Step-wise upsampling



Badrinarayanan et al. „SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation“. TPAMI 2016

SegNet

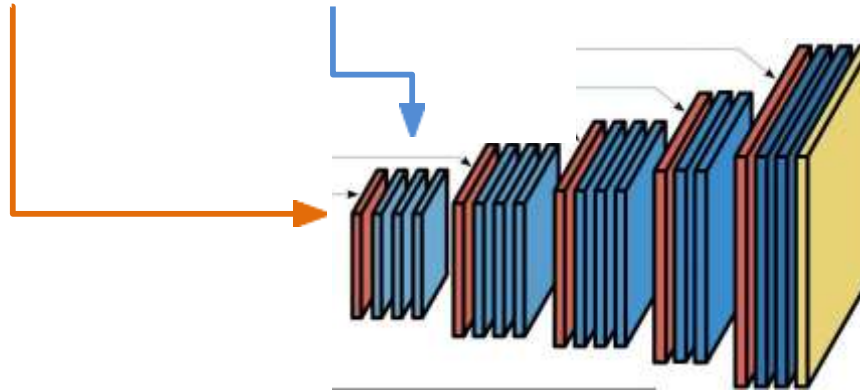
- Encoder: normal convolutional filters + pooling
- Decoder: Upsampling + convolutional filters



Badrinarayanan et al. „SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation“. TPAMI 2016

SegNet

- Encoder: normal convolutional filters + pooling
- Decoder: Upsampling + convolutional filters



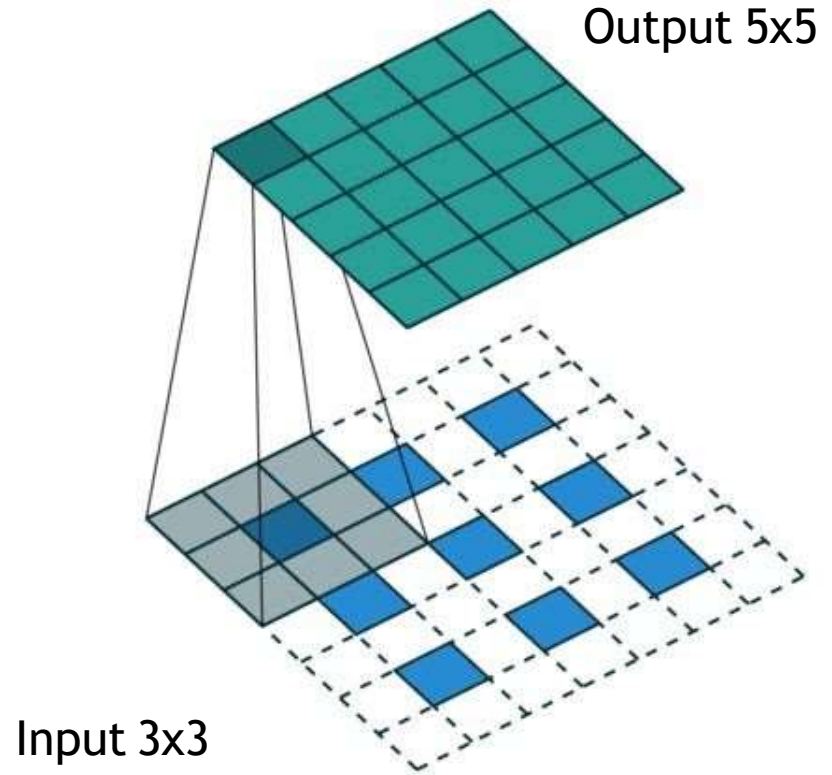
Badrinarayanan et al. „SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation“. TPAMI 2016

SegNet

- Encoder: normal convolutional filters + pooling
- Decoder: Upsampling + convolutional filters
- The convolutional filters in the decoder are learned using backprop and their goal is to refine the upsampling

Transposed convolution

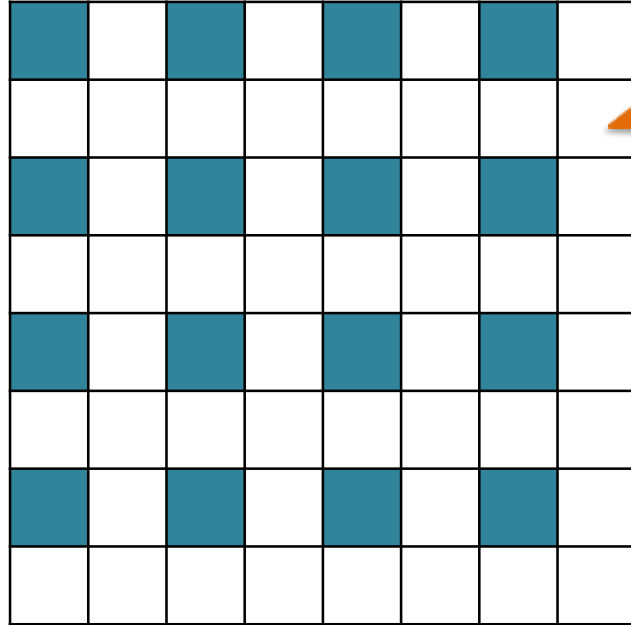
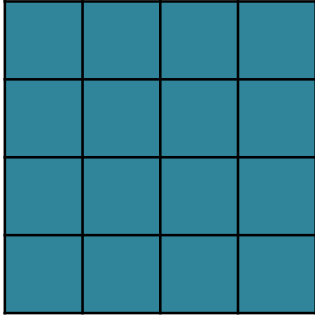
- Transposed convolution
 - Unpooling
 - Convolution filter (learned)
 - Also called up-convolution (never deconvolution)



Upsampling

Types of upsamplings

- 1. Interpolation



?

Types of upsamplings

- 1. Interpolation

Original image  x 10



Nearest neighbor interpolation

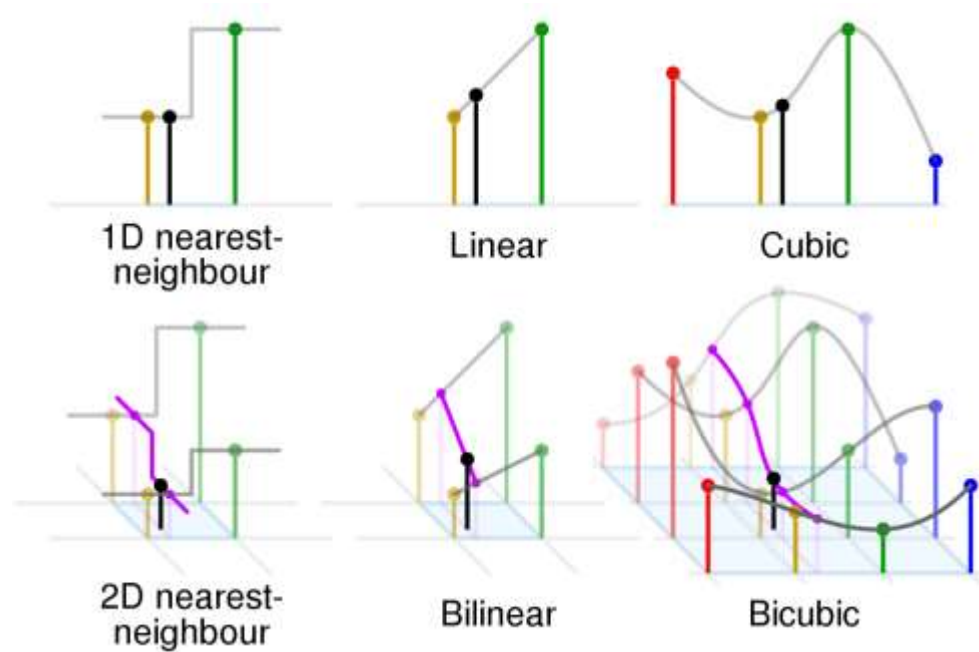


Bilinear interpolation



Bicubic interpolation

Types of Interpolations



Nearest just copies pixels, bilinear averages neighbors, and bicubic produces smoother transitions using more context.”

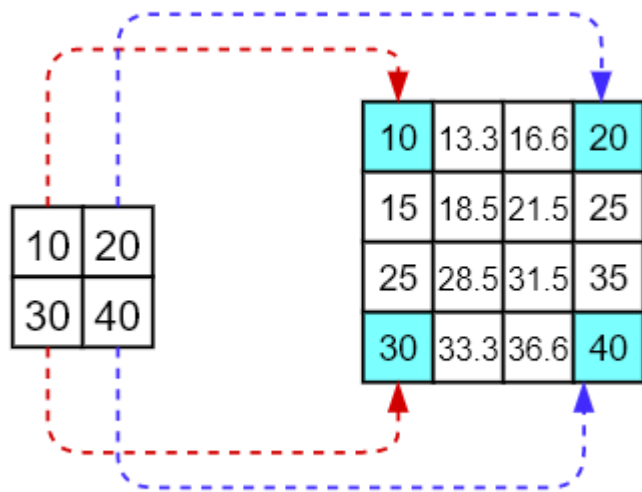
10	20
30	40

2x2

2x

10	10	20	20
10	10	20	20
30	30	40	40
30	30	40	40

4x4



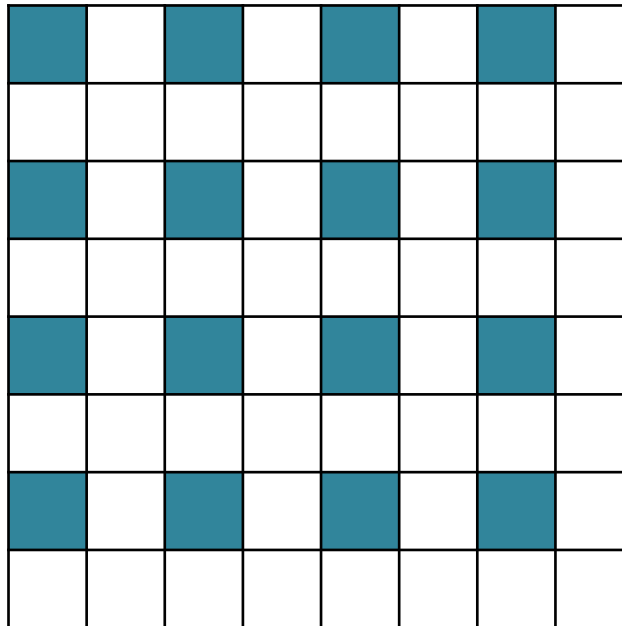
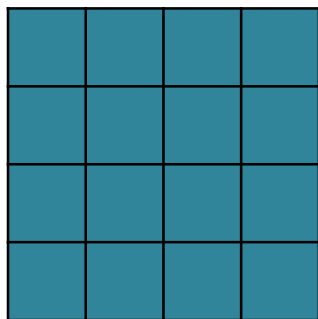
Types of upsamplings

- 1. Interpolation

 Few artifacts

Types of upsamplings

- 2. Fixed unpooling



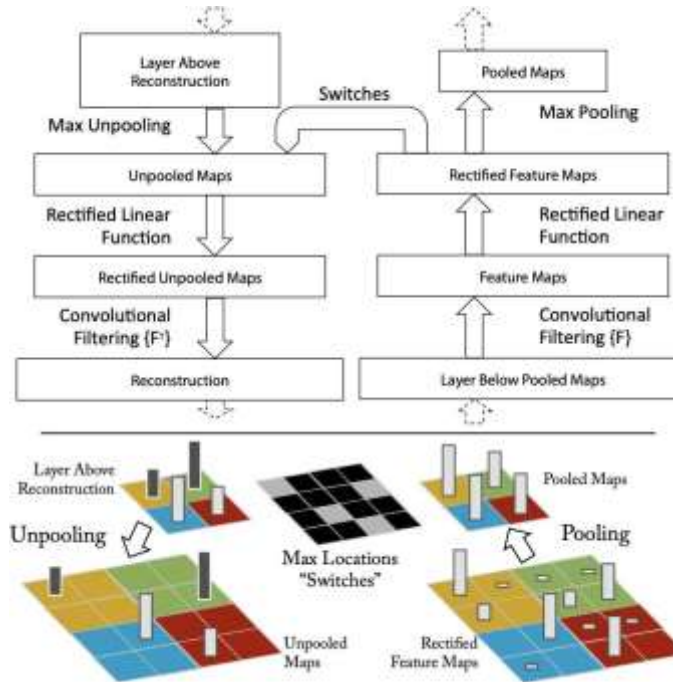
efficient

+ CONVS

A. Dosovitskiy, “Learning to Generate Chairs, Tables and Cars with Convolutional Networks“. TPAMI 2017

Types of upsamplings

- 3. Unpooling: “à la DeconvNet”



Keep the
locations where
the max came
from

Types of upsamplings

- 3. Unpooling: “à la DeconvNet”



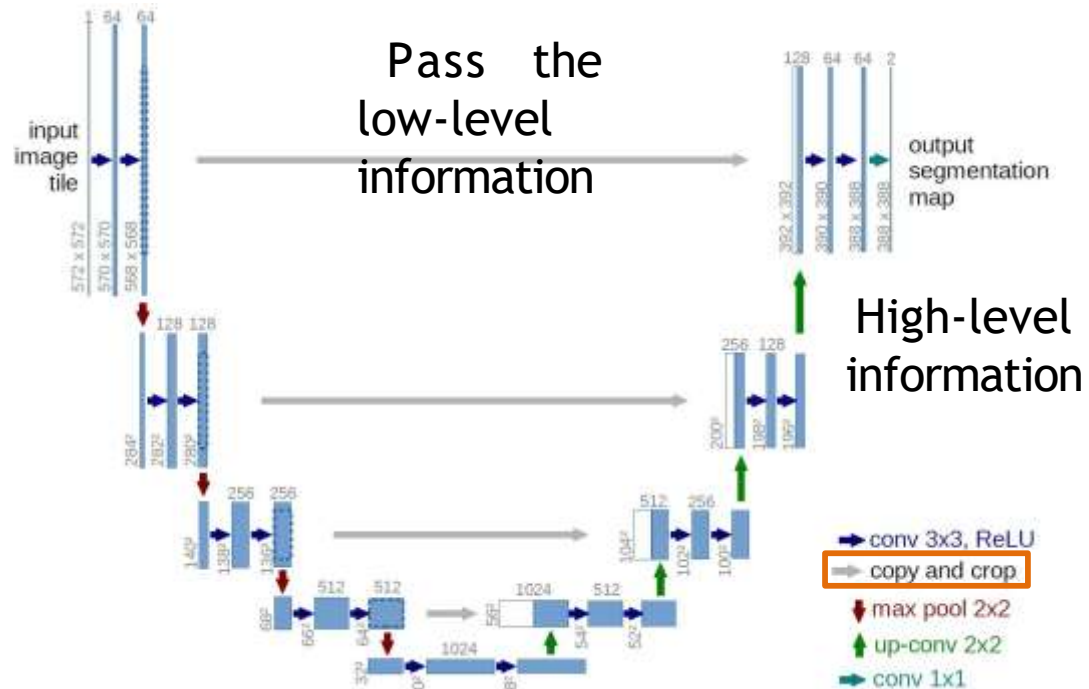
Keep the details of the structures

Skip connections (U-Net)

Skip Connections

- U-Net

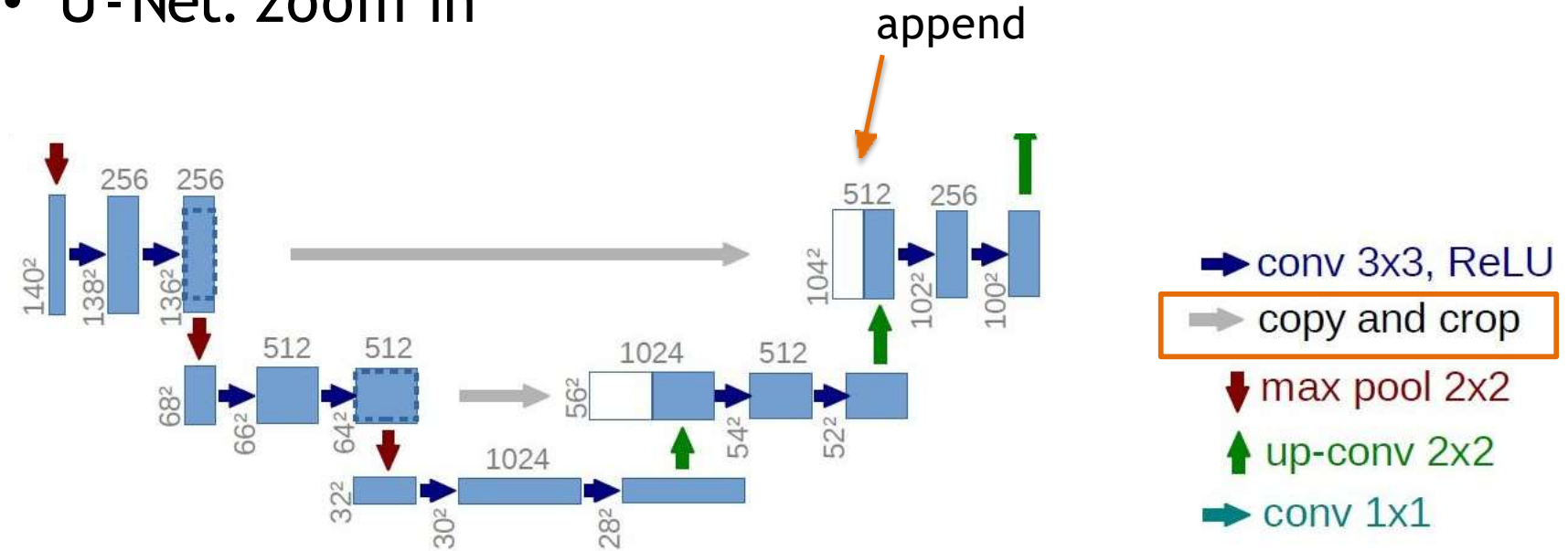
Recall ResNet



O. Ronneberger et al. "U-Net: Convolutional Networks for Biomedical Image Segmentation". MICCAI 2015

Skip Connections

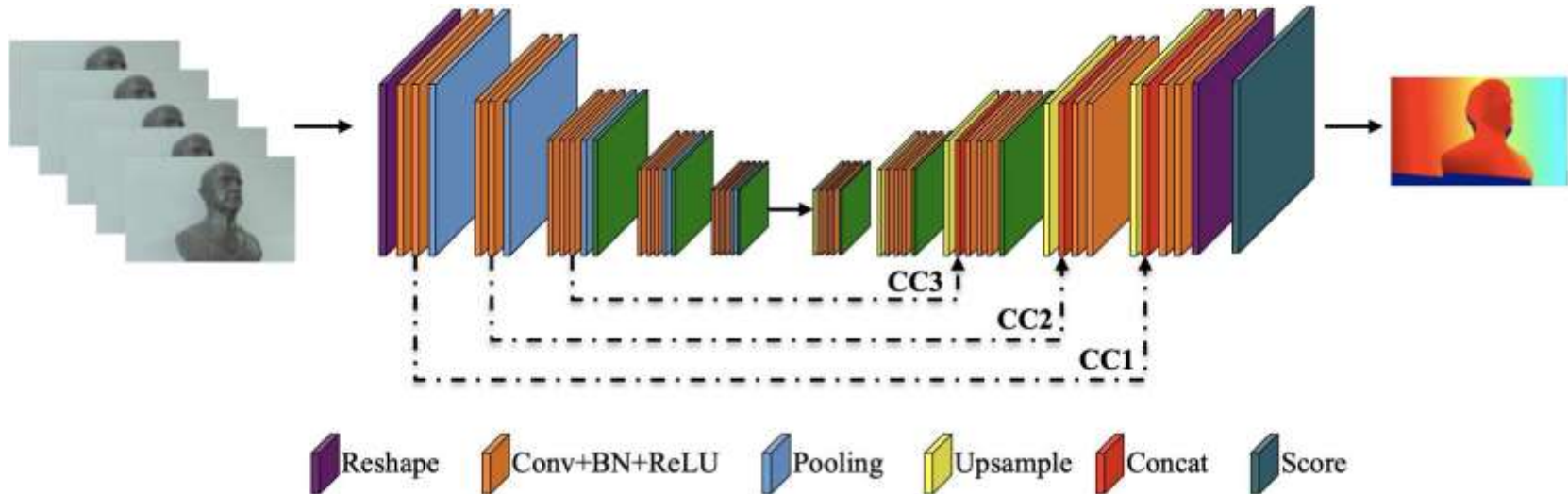
- U-Net: zoom in



O. Ronneberger et al. "U-Net: Convolutional Networks for Biomedical Image Segmentation". MICCAI 2015

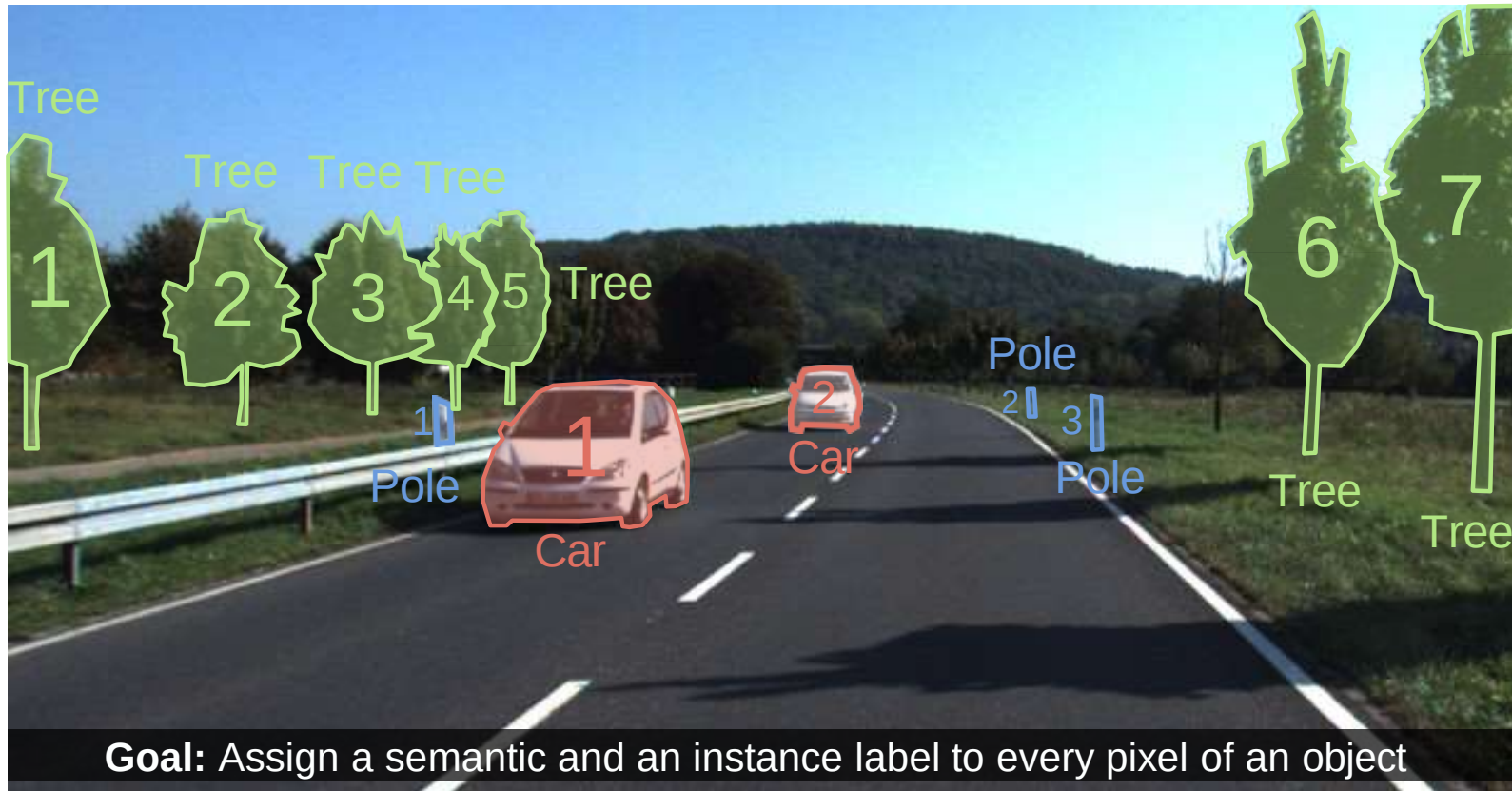
Skip Connections

- Concatenation connections

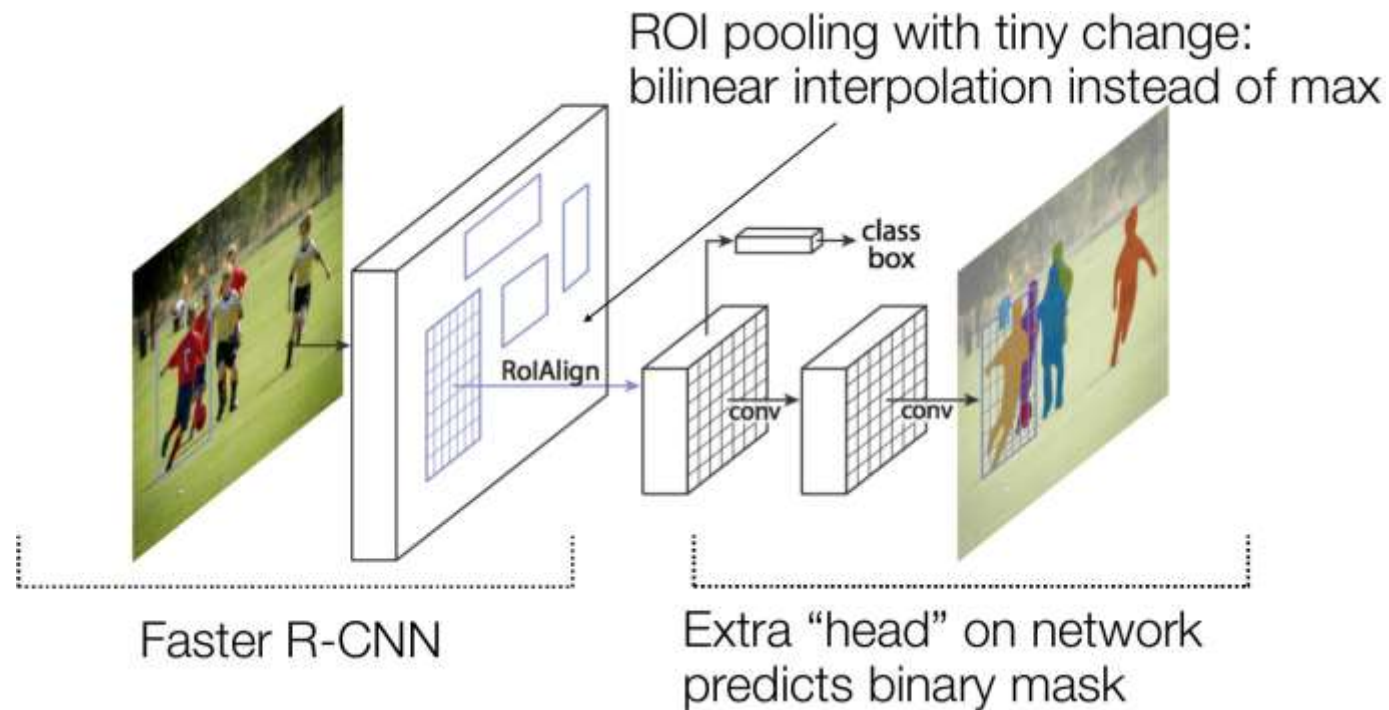


Instance Segmentation

Instance Segmentation

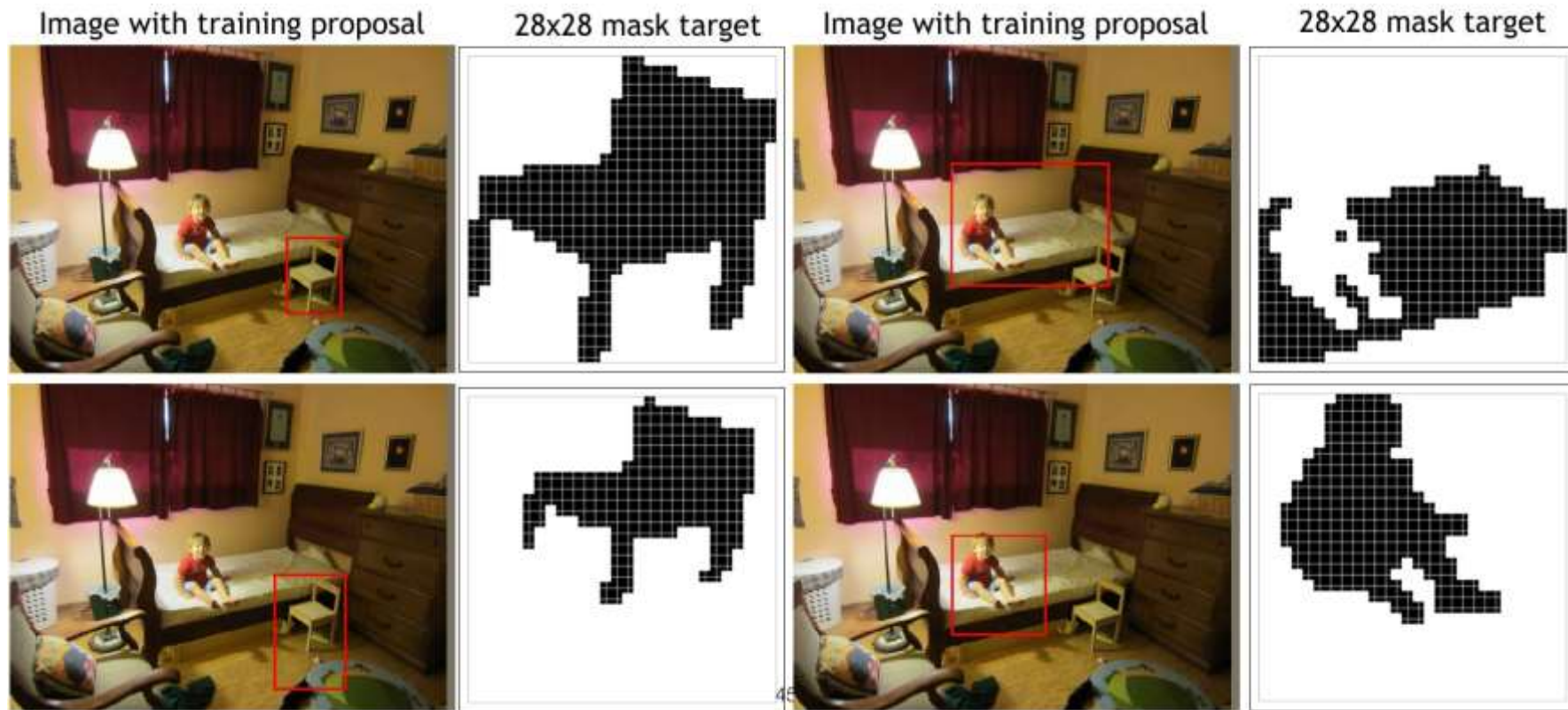


Mask R-CNN: Faster R-CNN for Instance Segmentation



- Predict binary instance segmentation (foreground vs. background) per detection

Mask R-CNN: Faster R-CNN for Instance Segmentation



Mask R-CNN: Faster R-CNN for Instance Segmentation



- Evaluation: Intersection-over-Union at mask level (not bounding box level)

Detectron2 for PyTorch



<https://github.com/facebookresearch/detectron2>

Datasets and metrics

Datasets

Pascal VOC
2012:

9993 natural
images
divided into
20 classes.

Cityscapes:

25K
urban-street
images
divided into
30 classes.

ADE20K:

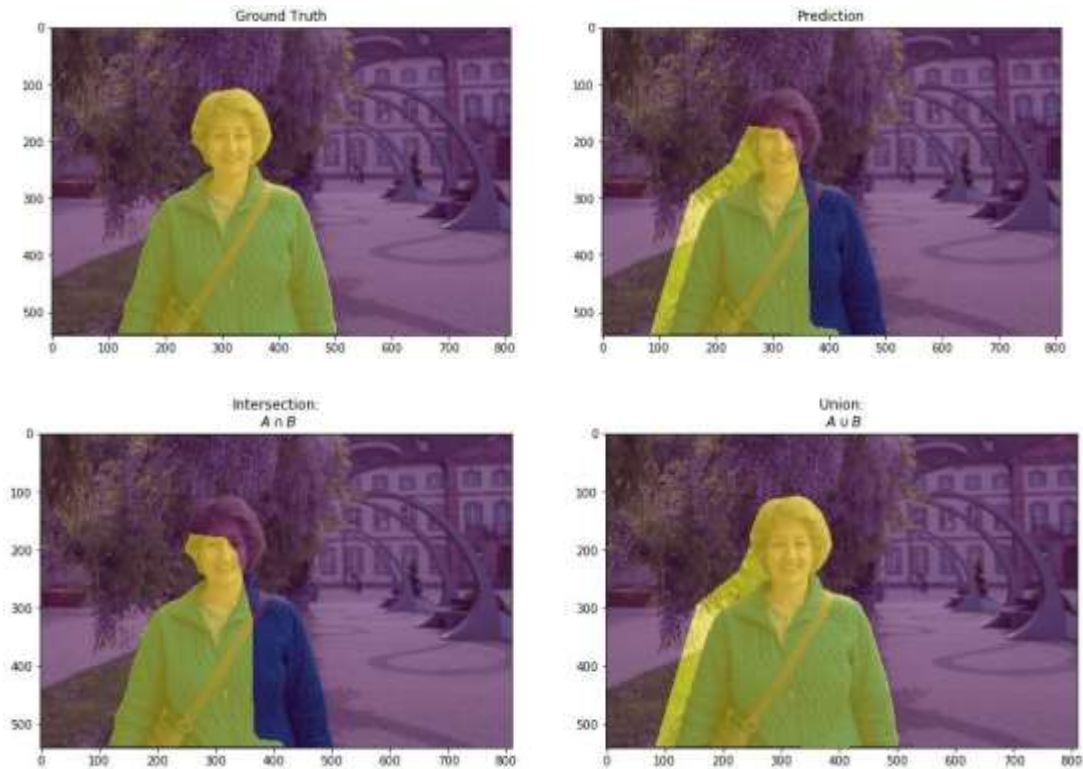
25K (20 stands
for 20K training)
scene-parsing
images divided
into
150 classes.

Mapillary
Vistas:

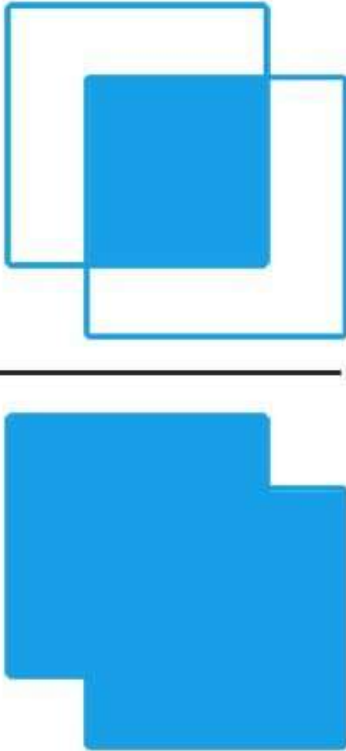
25K street
level images,
divided into
152 classes.

Models are often pre-trained in the large MS-COCO dataset,
before finetuned to the specific dataset.

Metrics: intersection over union (IoU)

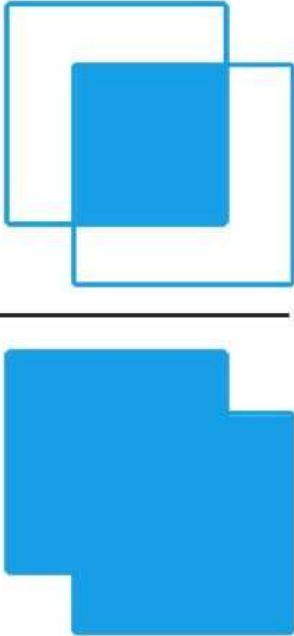


Metrics: intersection over union (IoU)

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


The diagram illustrates the IoU metric. It shows two overlapping rectangles: a yellow one and a blue one. The intersection is the overlapping area, and the union is the combined area of both rectangles. Below the formula, a solid blue shape represents the union of the two rectangles.

Metrics: mean intersection over union (mIoU)

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


MIoU simply computes the IoU for each class and then computes the mean of those values.

Another widely used metric is the pixel accuracy (ratio of pixels classified correctly).

Thank you for
listening